

Econ 2 - Lecture 11 - 5/6/26

Lecture Quiz #5 Released Today, due Monday, 5/11

Activity #3 This week $\rightarrow$ Top 3 of 5 count for 5% each

Today: Analyzing $Y = AE \Rightarrow$ Fiscal Policy, Chapters 4.3 & 5.1

Next Week: Deficits & Debt

Last Class: Found where $Y = AE$! Finally!

Y	C	I^P	G	Nx	AE	Chg. in Inventory
200	200	10	10	5	225	$200 - 225 = -25$
300	275	10	10	5	300	$\emptyset$
400	350	10	10	5	375	$400 - 375 = +25$
500	425	10	10	5	450	$500 - 450 = +50$
600	500	10	10	5	525	$600 - 525 = +75$

Equilibrium? Change in inventory? AC ? $C = AC + mpc(Y - T)$

$mpc =$ marginal propensity to consume $= \frac{\Delta C}{\Delta Y}$
 $\rightarrow$ slope of AE curve

Find where $Y = AE = C + I^P + G + Nx$

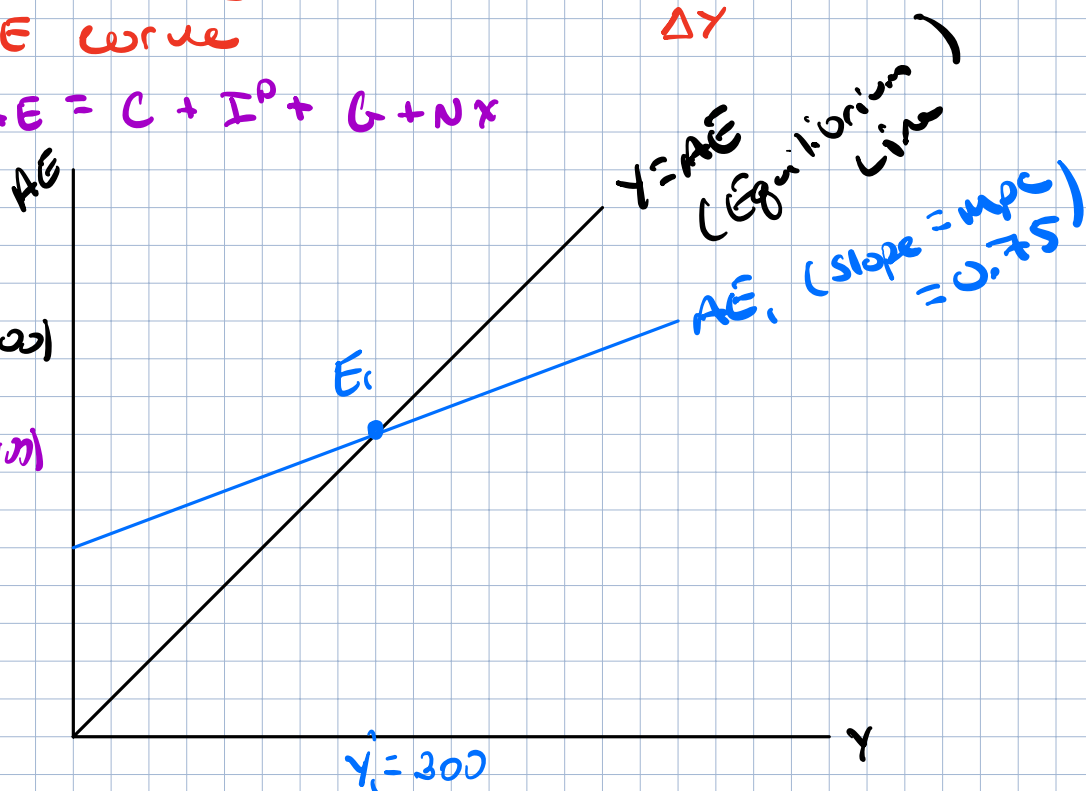
$T = 100$ in given table

$$C = AC + 0.75 \cdot (Y - 100)$$

$$200 = AC + 0.75(200 - 100)$$

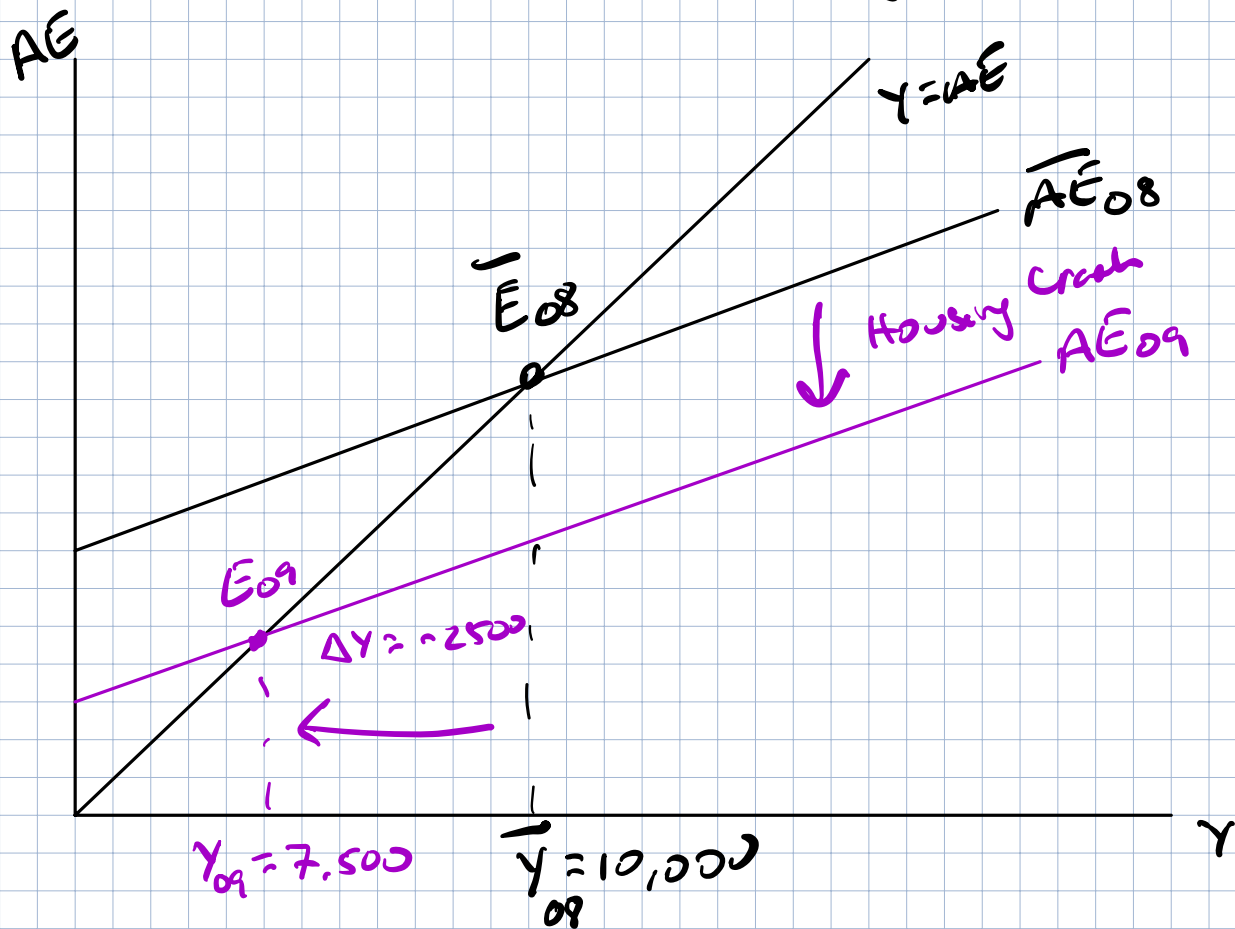
$$200 = AC + 75$$

$$AC = 125$$



Where next? Explain economic movements

Start at full-employment level of $Y = 10,000 = \bar{Y}$



Housing Crash $\rightarrow$ Reduction Wealth

Consumer Confidence fall $\rightarrow$ future concerns

$\rightarrow$ Reduction in Spending $\Rightarrow$ AE $\rightarrow$ Decrease in AD

$\rightarrow$ Investment Spending $\rightarrow$ Housing Developers? Decrease I^p

Graphically show decrease in AD, $I^p \rightarrow \downarrow$ in AE

$\Rightarrow$ Decrease in equilibrium $Y \rightarrow Y = 7,500 < 10,000$

$\hookrightarrow$ Unemployment rises $\Rightarrow$ Cyclical Unemployment

Not reaching the goal of full employment

Response to unemployment

Ask for help: Warren Buffett $\rightarrow$ increase spending!

$\hookrightarrow$ Buy some capital for his companies

$\bar{Y} = 10,000$, current $Y = 7,500$, estimate $mpc = 0.8$

How much should we ask for? Less than 2500.

Why? Round 1, Warren buy \$500 in machines from Bella

$\hookrightarrow$ Increase in IP of \$500

$\rightarrow$ Increase in Y of \$500

Buy mpc

Buy $mpc \times 500$
worth of oranges

$$0.8 \times 500 = \$400$$

from Penny



Consumption
Rises \$400

Buy $mpc \times 400$

Buy 0.8×400
 $= \$320$ in
Phones from
Olive

Increase in C, Y

Round 3
 $500 + 400 + 320 = 1220$

Round 2

Total Change
in $Y = 500 + 400 = 900$

Save $1 - mpc = mps$

$$\text{Save } (1 - 0.8) \times 500 = 0.2 \times 500 = \$100$$

Save 0.2×400

\$80
in savings

Go on for 50+ Rounds $\rightarrow$ Penny stamp of \$0.01

How much is our total increase in Y ?

Exactly 2500!

Each dollar spent in Round 1
Multiplied through economy 5 times!

$$\text{Expenditure Multiplier} = \frac{1}{1 - \text{MPC}}$$

$$\text{If } \text{MPC} = 0.8, \text{ Expenditure Multiplier} \\ = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5$$

$$\Delta Y = \frac{1}{1 - \text{MPC}} \Delta I^P, \quad \Delta I^P \text{ in Round 1} = 500 \\ \frac{1}{1 - \text{MPC}} = 5$$

$$\Delta Y = 5 \times 500 = 2500$$

Started at $Y_{09} = 7,500$, goal was $\bar{Y} = 10,000$

$$\text{After } \Delta I^P \rightarrow \text{new } Y = Y_{09} + \Delta Y = 7,500 + 2500 = 10,000$$

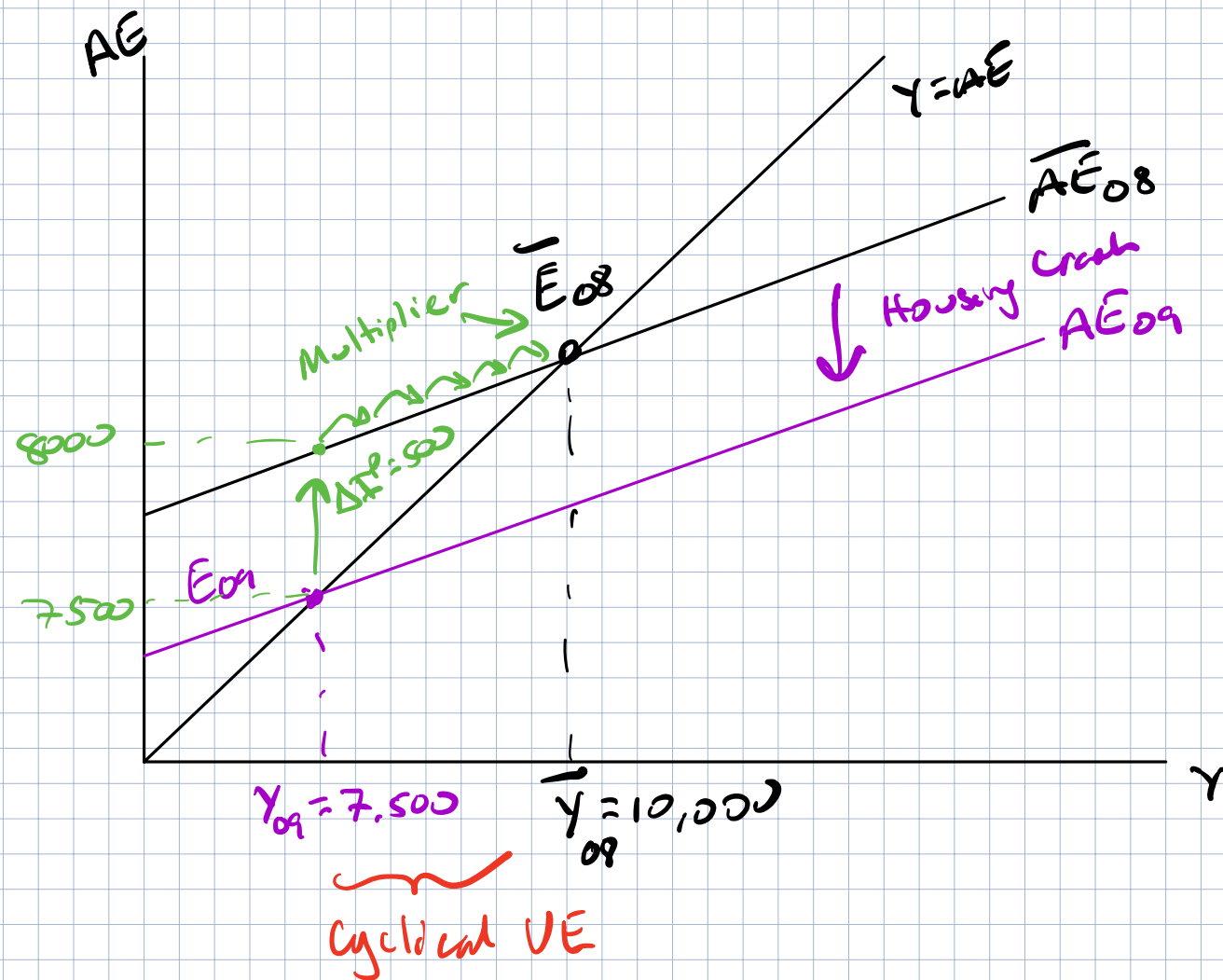
Note: If $\Delta G = 500$, $\Delta AC = 500$, $\Delta NX = 500$

$\rightarrow$ All have the same multiplier

$$\text{End up at } \Delta Y = 2500$$

Applies to I^P , G , NX , AC

Does not apply to taxes (T)



$mpc = 0.75$ Housing crash reduces AC, ΔY

$$\Delta Y = \frac{1}{1-mpc} \Delta AC$$

Housing crash $\rightarrow \Delta AC = -15000$

$$\Delta Y = \frac{1}{1-0.75} (-15000) = 4 \times -15000 = -60000$$

G

$$G' = G + \Delta G$$

$$AE' = C + 840$$

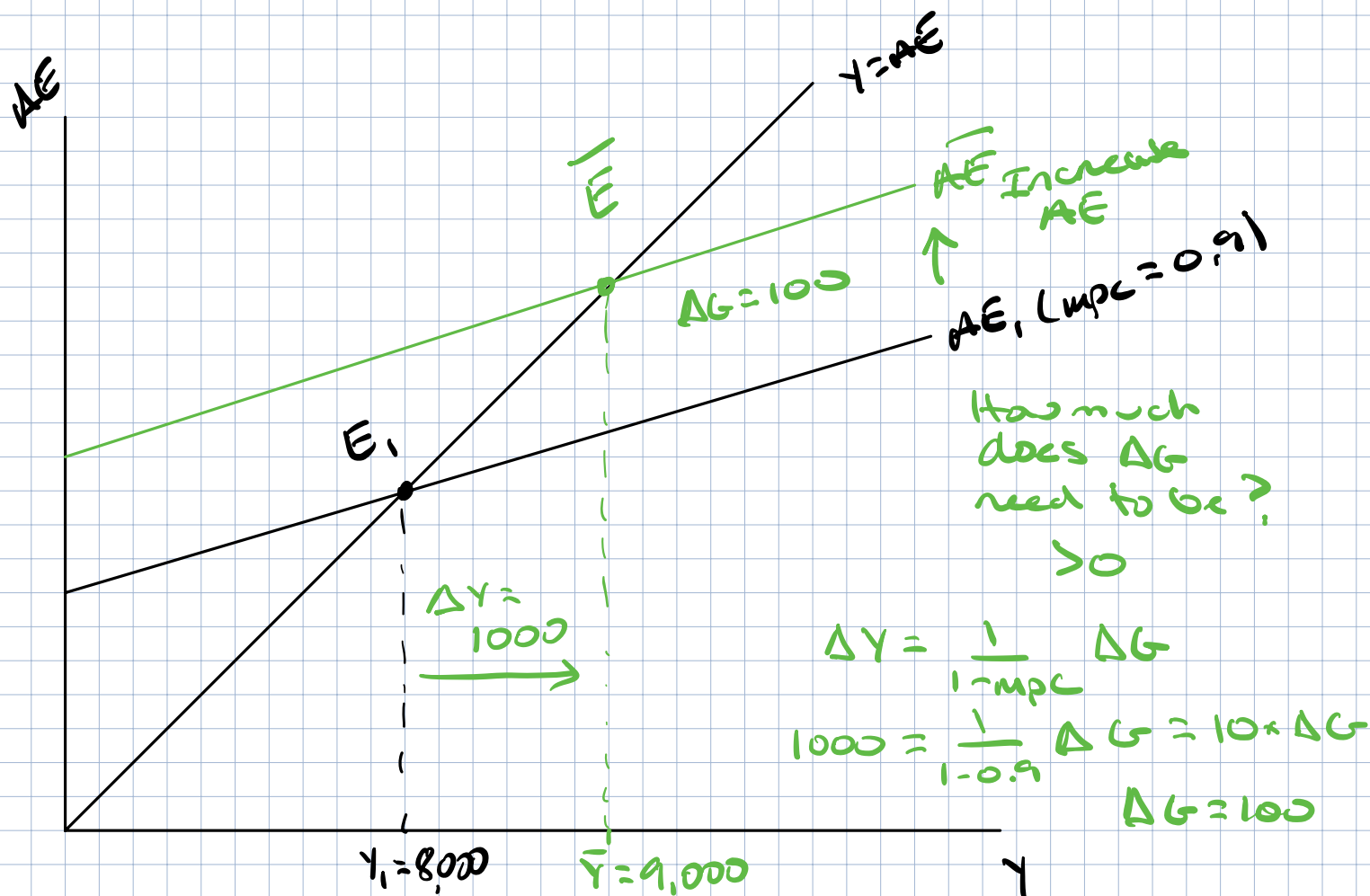
Real GDP (Y)	Consumption (C)	Planned Investment (I ^P)	Government Expenditures (G)	Net Exports (NX)	Aggregate Expenditures (AE)
5000	4560	500	200	300	5300
6000	5460	500	200	300	6200
7000	6360	500	200	300	7100
8000	7260	500	200	300	8000
9000	8160	500	200	300	8900
10000	9060	500	200	300	9800

What is the level of $Y = AE$?

$$MPC = \frac{\Delta C}{\Delta Y} = \frac{6360 - 5460}{7000 - 6000} = \frac{900}{1000} = 0.9$$

$$I^P + G + NX = 500 + 200 + 40 = 740$$

At $Y = 8000$, $AE = 8000 \rightarrow$ Equilibrium, $\bar{Y} = 9,000$, ΔG ?



Recap: Find Short-Run Equilibrium ($Y = AE$)

Table, graph, story, etc.

Incorporate economic shocks into model

Housing Crash $\rightarrow \downarrow AC, \downarrow I^P \downarrow$

$$\Delta Y = \frac{1}{1 - mpc} (\Delta AC + \Delta I^P)$$

More Recent: Covid-19 Pandemic

In 2019: Full-Employment Level of $\bar{Y} \Rightarrow UE = 3.5\%$
in Dec 2019

